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School of Mathematics

Bayesian Theory Notes

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About these Notes

These notes are based on the course **Bayes Theory**, which follows the framework of *Bayesian Statistical Methods* by **Brian J. Reich & Sujit K. Ghosh**. This course is designed as an advanced undergraduate or foundational postgraduate-level module.

I attended this module at **The University of Edinburgh** during the **2022/23 academic year**. Some time has passed since completing this course, yet I felt compelled to consolidate my original notes into this comprehensive reference to support the wider academic community in their learning and research. If you identify any errors or have suggestions for improvement while reviewing these notes, please feel free to contact me via email at: chenyz9811@gmail.com.

1 Probability Background

Probability theory provides the mathematical foundation for statistical analysis. In probability, we usually assume that a mathematical model for a random process is already known, and we use that model to compute the probabilities of events. For example, if a fair coin is tossed five times, probability theory can be used to calculate the probability of observing five heads.

Statistics works in the opposite direction. In statistics, we observe data and use those data to learn about the underlying process that generated them. For example, if we observe five heads in a row, we may ask whether the coin is fair or whether there is evidence that it is biased.

Therefore, probability and statistics are closely related but have different purposes:

Probability: model $\rightarrow$ data behaviour

Statistics: data $\rightarrow$ model inference

In other words, probability asks what data we should expect under a given model, whereas statistics uses observed data to make statements about the model.

1.1 Events and Probabilities

An **event** is a set of possible outcomes from a random experiment. Events are often denoted by capital letters such as

A

The probability of an event A is written as

$Prob(A)$

For example, if A is the event that a coin lands heads, then $Prob(A)$ is the probability of observing heads.

A random variable is usually denoted by a capital letter, such as

X

A particular realised value of that random variable is usually denoted by a lower-case letter, such as

x

For example,

$Prob(X = x)$

means the probability that the random variable X takes the specific value x .

If X is a continuous or interval-valued random variable, we may also write probabilities such as

$Prob(X \in [a, b])$

which means the probability that X lies in the interval $[a,b]$.

1.2 Interpretations of Probability

There are two common interpretations of probability: the frequentist interpretation and the subjective interpretation.

Under the **frequentist interpretation**, the probability of an event is understood as its long-run relative frequency. If the same experiment were repeated many times, the proportion of times that event A occurred would eventually approach $Pr(A)$.

For example, if a fair coin is tossed many times, the proportion of heads should approach

$$0.5$$

Under the subjective interpretation, probability represents a degree of belief about whether an event will occur. This interpretation is closely associated with Bayesian statistics. For example, an individual may assign a probability to the event that it will rain tomorrow based on available information, experience and judgement.

These two interpretations lead naturally to two major approaches in statistics:

Frequentist statistics

and

Bayesian statistics

In Bayesian analysis, probability is often used to represent uncertainty about unknown quantities, including model parameters. This is why probability theory is essential for understanding Bayesian statistical methods.

1.3 Key Distinction Between Probability and Statistics

A useful way to remember the difference is:

Field	Starting Point	Main Question
Probability	A known model	What data or outcomes should we expect?
Statistics	Observed data	What can we infer about the model or process?

For example: In probability, we might assume that a coin is fair and calculate

$$\text{Prob}(\text{five heads in five tosses})$$

In statistics, we might observe five heads in five tosses and ask whether this provides evidence that the coin is biased.

Key Takeaways Probability theory provides the mathematical language for describing uncertainty. An event is denoted by A , and its probability is written as

$$\text{Prob}(A)$$

A random variable is usually written as a capital letter, such as X , while its realised value is written as a lower-case letter, such as x .

The frequentist interpretation views probability as long-run relative frequency.

The subjective interpretation views probability as a degree of belief.

Probability starts with a model and studies possible outcomes.

Statistics starts with data and makes inference about the model or data-generating process.

Bayesian statistics uses probability to represent uncertainty about unknown quantities.

2 Univariate Discrete Distributions

2.1 Random Variables and Support

A random variable X assigns numerical values to the outcomes of a random experiment. The support of X , usually denoted by S , is the set of all values that X can take with probability one.

For example, if X is the number of successes in n trials, then the possible values are

$$S = \{0, 1, \dots, n\}.$$

This is because the number of successes cannot be smaller than 0 or greater than n .

A random variable is called discrete if its support is countable. The support may be finite, such as

$$S = \{0, 1, \dots, n\},$$

or countably infinite, such as

$$S = \{0, 1, 2, \dots\},$$

A random variable is called continuous if its possible values form an interval or another uncountable set. For example, the amount of rainfall on a given day may take any non-negative real value, so its support could be

$$S = [0, \infty)$$

2.2 Probability Mass Function

For a discrete random variable, the probability mass function, or PMF, assigns a probability to each value in the support of X . It is defined as

$$f(x) = \text{Prob}(X = x).$$

A function $f(x)$ is a valid PMF if it satisfies two conditions:

$$f(x) \geq 0 \text{ for all } x \in S$$

and

$$\sum_{x \in S} f(x) = 1.$$

The first condition states that probabilities cannot be negative. The second condition states that the probabilities of all possible outcomes must add up to one. If S' is a subset of the support S , then the probability that X falls in S' is

$$\text{Prob}(X \in S') = \sum_{x \in S'} f(x)$$

For example, if x_1 and x_2 are two possible values of X , then

$$\text{Prob}(X = x_1 \text{ or } X = x_2) = f(x_1) + f(x_2)$$

2.3 Cumulative Distribution Function

The cumulative distribution function, or **CDF**, of a random variable X is defined as

$$F(x) = \text{Prob}(X \leq x)$$

For a discrete random variable, the CDF is obtained by summing the PMF over all values less than or equal to x :

$$F(x) = \sum_{x \leq c} f(c)$$

The CDF therefore gives the probability that the random variable takes a value no greater than x .

2.4 Mean and Variance

The expected value, or mean, of a discrete random variable is

$$E(X) = \sum_{x \in S} x f(x)$$

The mean describes the centre of the distribution.

The variance measures the spread of the distribution around its mean. It is defined as

$$Var(X) = E[X - E(X)]^2$$

For a discrete random variable, this becomes

$$Var(X) = \sum_{x \in S} [x - E(X)]^2 f(x)$$

A useful equivalent formula is

$$Var(X) = E(X^2) - [E(X)]^2$$

where

$$E(X^2) = \sum_{x \in S} x^2 f(x)$$

The standard deviation is the square root of the variance:

$$SD(x) = \sqrt{Var(X)}$$

It is often easier to interpret than the variance because it is measured on the same scale as X .

2.5 Samples, Statistics and Sampling Distributions

In statistical analysis, we often observe a sample from a population. A sample may be written as

$$X_1, \dots, X_n$$

where each X_i is a random variable from the same population distribution.

The sample mean is

$$\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i$$

It is important to distinguish between the population mean $E(X)$ and the sample mean $\bar{X}$. The population mean is a fixed feature of the distribution, whereas the sample mean depends on the particular sample observed.

A **statistic** is any function of the sample. For example, $\bar{X}$ is a statistic. Since a statistic depends on random data, it is itself a random variable. The probability distribution of a statistic is called its **sampling distribution**.

2.6 Parametric Statistical Models

A **parametric model** assumes that the distribution of X belongs to a known family of distributions, but that one or more parameters are unknown.

We often write the PMF as

$$f(x|\theta)$$

which means the probability of observing $X = x$ given the parameter θ .

The vertical line in $f(x | \theta)$ should be read as “given”. It is not division.

The advantage of a parametric model is that the whole distribution can often be described using only a small number of parameters. Once these parameters are known or estimated, the probabilities, mean, variance and other properties of the distribution can be determined.

2.7 Common Discrete Distributions

2.7.1 Bernoulli Distribution

A Bernoulli random variable models a single trial with two possible outcomes: success and failure. We usually code success as 1 and failure as 0. If

$$X \in \{0, 1\}$$

and

$$\text{Prob}(X = 1 \mid \theta) = \theta$$

then

$$\text{Prob}(X = 0 \mid \theta) = 1 - \theta$$

We write

$$X \sim \text{Bernoulli}(\theta)$$

where

$$0 \leq \theta \leq 1$$

The PMF is

$$f(x \mid \theta) = \theta^x (1 - \theta)^{1-x}, x = 0, 1$$

This compact formula works because : when $x = 1$:

$$f(1 \mid \theta) = \theta^1 (1 - \theta)^0 = \theta$$

and when $x = 0$:

$$f(0 \mid \theta) = \theta^0 (1 - \theta)^1 = 1 - \theta$$

The mean of a Bernoulli random variable is

$$E(X \mid \theta) = \sum_{x=0}^1 x f(x \mid \theta) = f(1 \mid \theta) = \theta$$

The variance is $\theta(1 - \theta)$

$$\begin{aligned} \text{Var}(X \mid \theta) &= \sum_{x=0}^1 [x - E(X \mid \theta)]^2 f(x \mid \theta) \\ &= [0 - \theta]^2 \theta^0 (1 - \theta)^{1-0} + [1 - \theta]^2 \theta^{1-0} (1 - \theta)^0 \\ &= \theta^2 - \theta^3 + \theta - 2\theta^2 + \theta^3 \\ &= \theta - \theta^2 \\ &= \theta(1 - \theta) \quad (\text{End Proof}) \end{aligned} \tag{2.1}$$

2.7.2 Binomial Distribution

The binomial distribution generalises the Bernoulli distribution from one trial to n independent trials. Suppose

$$X_1, \dots, X_n$$

are independent Bernoulli random variables with success probability θ . That is,

$$X_i \sim \text{Bernoulli}(\theta), i = 1, \dots, n.$$

Let

$$X = \sum_{i=1}^n X_i$$

Then X is the total number of successes in n independent trials, and

$$X \sim \text{Binomial}(n, \theta)$$

The support is

$$S = \{0, 1, \dots, n\}$$

The PMF is

$$f(x | \theta) = \binom{n}{x} \theta^x (1 - \theta)^{n-x}, x = 0, 1, \dots, n.$$

Here,

$$\binom{n}{x}$$

is the binomial coefficient, which counts the number of ways to choose x successes from n trials.

The mean is

$$E(X | \theta) = n\theta$$

The variance is

$$\text{Var}(X | \theta) = n\theta(1 - \theta)$$

This follows from the fact that X is the sum of n independent Bernoulli random variables. Since each Bernoulli random variable has mean θ ,

$$E(X | \theta) = E\left(\sum_{i=1}^n X_i | \theta\right) = \sum_{i=1}^n E(X_i | \theta) = n\theta$$

Since the X_i 's are independent,

$$\text{Var}(X | \theta) = \sum_{i=1}^n \text{Var}(X_i | \theta) = \sum_{i=1}^n \theta(1 - \theta) = n\theta(1 - \theta)$$

2.7.3 Poisson Distribution

The Poisson distribution is commonly used to model count data, especially the number of events occurring in a fixed interval of time or space.

Examples include:

- the number of calls received by a call centre in one hour;
- the number of accidents at a junction in one month;
- the number of emails received in a day;
- the number of events occurring in a fixed observation period.

If

$$X \sim \text{Poisson}(\theta)$$

then the support is

$$S = \{0, 1, 2, \dots\}$$

where

$$\theta \geq 0,$$

The PMF is

$$f(x | \theta) = \frac{e^{-\theta} \theta^x}{x!}, x = 0, 1, 2, \dots$$

The mean and variance of a Poisson random variable are both equal to θ :

$$E(X \mid \theta) = \theta$$

and

$$Var(X \mid \theta) = \theta$$

This equality between the mean and variance is an important property of the Poisson distribution.

Distribution	Support	PMF	Mean	Variance
Bernoulli(θ)	$\{0, 1\}$	$\theta^x(1 - \theta)^{1-x}$	θ	$\theta(1 - \theta)$
Binomial(n, θ)	$\{0, 1, \dots, n\}$	$\binom{n}{x}\theta^x(1 - \theta)^{n-x}$	$n\theta$	$n\theta(1 - \theta)$
Poisson(θ)	$\{0, 1, 2, \dots\}$	$\frac{e^{-\theta}\theta^x}{x!}$	θ	θ

Table 1: Summary of Common Discrete Distributions

2.7.4 Key Takeaways

A discrete random variable has a countable support.

The PMF gives the probability of each possible value:

$$f(x) = Prob(X = x)$$

A valid PMF must be non-negative and sum to one.

The CDF gives cumulative probability:

$$F(x) = Prob(X \leq x).$$

The mean and variance of a discrete random variable can be calculated from its PMF:

$$E(X) = \sum_{x \in S} xf(s) \quad Var(X) = \sum_{x \in S} [x - E(X)]^2 f(x)$$

The Bernoulli distribution models one binary trial.

The binomial distribution models the number of successes in n independent Bernoulli trials.

The Poisson distribution models counts of events in a fixed interval.

In parametric statistics, we write

$$f(x \mid \theta)$$

to show that the distribution depends on an unknown parameter θ . The aim of statistical inference is often to estimate this parameter from data.